

RWANDA TVET BOARD
COMPETENCY-BASED ASSESSMENT
[FORMATIVE ASSESSMENT]

Program: SOFTWARE DEVELOPMENT | **Level:** 5 | **Total Marks:** 10

Module: Machine Learning Application | **Date:** 2026-05-25 | **Time:** 3 Hours

Name: _____ Index No: _____ Class: _____

SECTION A: TRUE / FALSE

Answer True (T) or False (F).

- 1 Supervised Learning has no role when applying Data Pre-processing. (1 marks)

Answer: **True** **False**

- 2 Linear Regression is a key concept when applying Data Pre-processing. (1 marks)

Answer: **True** **False**

SECTION B: MULTIPLE CHOICE

Choose ONE correct answer and circle the letter.

- 3 What is the significance of Unsupervised Learning when applying Data Pre-processing? (1 marks)

- A. Deep Q-Networks (DQN)
- B. Unsupervised Learning
- C. Model Accuracy
- D. Data cleaning

Answer: _____

- 4 What is the role of Reinforcement Learning when applying Data Pre-processing? (1 marks)

- A. Unsupervised Learning
- B. Loss function
- C. Computational power
- D. Reinforcement Learning

Answer: _____

- 5 Which approach to Training dataset is most effective when applying Data Pre-processing? (1 marks)

- A. Target label
- B. Training dataset
- C. NumPy array
- D. Linear regression

Answer: _____

SECTION C: MATCHING

Match each term in Column A with Column B.

- 6 Match each term in Column A with its correct description in Column B. (1 marks)

Column A (Terms)	Column B (Descriptions)
1. Machine learning life cycle	A. is a cyclic process to build an efficient machine learning project.
2. Speech recognition	B. is a process of converting voice instructions into text, and it is also
3. Machine learning	C. is an evolving field with ongoing research, offering opportunities for
4. Python	D. is a popular high-level, general-use programming language. Python is a

Answers: 1.____ 2.____ 3.____ 4.____

SECTION D: OPEN-ENDED

Answer ALL questions. Write in the spaces provided.

7 Explain the role of Test dataset when applying Data Pre-processing. **(1 marks)**

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8 Design a solution that uses ML Algorithm to applying Data Pre-processing. Explain your design decisions. **(2 marks)**

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9 Analyze the importance of Cross-validation in applying Data Pre-processing. Identify its advantages and limitations. **(2 marks)**

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1 Define Logistic Regression and explain its importance in applying Data Pre-processing. **(2 marks)**

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